

Ian M. Nesbitt

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National Center for Ecological Analysis and Synthesis, U.C. Santa Barbara

DataONE Federation Software and Network Operations

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Education

2017–2022 **Master's Student in Earth and Climate Sciences**, *School of Earth and Climate Sciences, College of Natural Sciences, Forestry, and Agriculture, University of Maine, Orono, ME, USA*

GPA: 3.97

Thesis: Using diverse methods to determine the volume and stratigraphy of sediment in a formerly glaciated lake in central Maine, USA

Advisor: [Seth W. Campbell](#), *Co-advisor:* [Sean M.C. Smith](#)

Associated Projects: [readgssi](#), a Python tool to read and process radar data; [SeidarT](#), a seismic and radar survey modeling toolbox

Addendum: I completed all coursework and research for this degree, but discontinued the program in 2022 prior to publishing my thesis to pursue full-time work at NCEAS.

2009–2013 **Bachelor of Arts in Geosciences with Honors**, *Williams College, Williamstown, MA, USA*

Thesis: A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range

Advisor: [David P. Dethier](#)

2006–2009 **Degree with Honors**, *Holderness School, Holderness, NH, USA*

Awards and Scholarships

2021 Golden Key (invited)

2020 US Antarctica Service Medal

2013 Sigma Xi induction

2012 Williams College Class of 1960 Scholar in Geosciences

Volunteer Experience

Since 2025 Co-Chair, Earth Science Information Partners ([ESIP](#)) [schema.org Cluster](#) with [Colin Smith](#)

Since 2022 Nordic Ski Coach, Mt. Greylock Regional School

2018–2023 Co-President, Williams College Class of 2013

2017–2018 Co-Director of Reunion Planning, Williams College Class of 2013

Work Experience

- Since 2022 **Research Software Engineer**, *National Center for Ecological Analysis and Synthesis (NCEAS)*, University of California Santa Barbara, Santa Barbara, CA, USA (*remote*)
Duties: Python and Java development of open-source computing tools to interpret, quality-check, archive, and serve large quantities of research science data. Working on **DataONE Federation** metadata interoperability and data processing tools. DataONE Network Operations duties include monitoring and troubleshooting of **DataONE** member nodes and coordinating with the community to ensure the health of the network.
Supervisor: **Matt Jones**
- 2022 **Geospatial Engineering Consultant**, *Muddy Water Initiative*, Boston, MA, USA
Duties: Geospatial database management, design of SQL joins and relates, QA/QC and digitization of field data, presentation-quality map design, client and stakeholder interaction for **Caroline Reeves**.
- 2021 **Geophysicist**, *International Thwaites Glacier Collaboration, Project GHC*, Orono, ME, USA
Duties: Processed ground-penetrating radar (GPR) survey data from the Hudson Mountains, West Antarctica, in collaboration with **John Woodward**, **Joanne S. Johnson**, **Brent Goehring**, and **Seth W. Campbell**.
- 2021 **Research Software Developer**, *University of Maine*, Orono, ME, USA
Duties: Tested and debugged code for geophysical modeling software **SeidarT**, developed a multi-platform install script for **Christopher C. Gerbi** and **Steven P. Bernsen**.
Projects: <https://github.com/UMainedynamics/seidart>
- 2019–2020 **Lead Geophysicist**, *Allan Hills Antarctic Expedition I-165-M, Princeton University - NSF Award Number 1744993*, Allan Hills Field Camp, East Antarctica
Duties: Collected and processed 100 km of ground-penetrating radar (GPR) survey data from a blue ice area on the eastern side of the Trans-Antarctic Mountains. Deep field survival, data collection, processing, use of snow machines in harsh conditions.
- 2018–2020 **Lead Scientist**, *Raspberry Shake, S.A.*, Boquete, Chiriqui, Panama
Duties: Technical support, software development, QA/QC of seismic instruments, development and QA/QC of educational materials.
Projects: <https://github.com/raspishake/rsudp>, <https://edu.raspberrysshake.org>
- 2014–2017 **Geophysical Scientist**, *e4sciences LLC*, Newtown, CT, USA
Duties: Led field team for many types of geophysical survey including Mobile LiDAR, ground-penetrating radar, high-precision GPS, multibeam echosounding, sub-bottom seismic, sidescan sonar. Field interpretation, and on-the-ground operations decision-making including boat handling.
Projects: Created survey-to-delivery workflow for 3D LiDAR end-user deliverables.

- 2013–2014 **Assistant Nordic Ski Coach, St. Michael's College, Colchester, VT, USA**
Duties: Created and supervised individualized training plans for athletes, technique instruction, race day logistics including ski preparation.

Teaching

- 2019 **Ground-Penetrating Radar Theory and Applications to Geological Mapping – SC2, Northeastern Section Meeting, Geological Society of America, Portland, ME, USA**
Seth W. Campbell and Ian M. Nesbitt (filling in for Steven A. Arcone) taught short course at NEGSA annual meeting, covering radar application, data collection, processing, and interpretation.
- 2018 **Global Environmental Change – ERS 201, University of Maine, Orono, ME, USA**
Karl J. Kreutz and Ian M. Nesbitt instructed students in accepted and debated climate theory from an earth systems perspective.
- 2017 **Earth Systems – ERS 200, University of Maine, Orono, ME, USA**
Aaron E. Putnam, Sean M.C. Smith, Peter O. Koons, and Ian M. Nesbitt instructed students in earth systems theory.
- 2012–2013 **GIS and Remote Sensing – GEOS 214, Williams College, Williamstown, MA, USA**
David P. Dethier and Ian M. Nesbitt instructed students in traditional and emerging geomorphologic theory, methods, and practice.
- 2012 **Geomorphology – GEOS 201, Williams College, Williamstown, MA, USA**
David P. Dethier and Ian M. Nesbitt instructed students in geographical information systems (GIS) theory, technology and software, and advised in the application of these methods via end-of-term projects.

Peer-Reviewed Publications

- in review **The A-KBS Arctic Knowledge Based System: Science Gateway Integration for Exascale Arctic Data Processing and Geospatial Feature Prediction, ESS Open Archive**
Meisam Shayeghmoradi, Andrew Wilcox, Sheridan Parker, Aaron Bergstrom, Ian M. Nesbitt, Naima Kaabouch, and Timothy Pasch
preprint: [10.22541/essoar.177292734.41397214/v1](https://doi.org/10.22541/essoar.177292734.41397214/v1)
- 2026 **Broadly stable atmospheric CO₂ and CH₄ levels over the past 3 million years, Nature**
J. Marks-Peterson, Sarah Shackleton, John Higgins et al.
doi: [10.1038/s41586-025-10032-y](https://doi.org/10.1038/s41586-025-10032-y)

- 2026 **The arctic knowledge-based system: Science gateway integration for petascale arctic data processing and geospatial feature prediction**, *Applied Computing and Geosciences*
Andrew Wilcox, Meisam Shayeghmoradi, Stephen Miller, [Ian M. Nesbitt](#), Saisri Pogalla, Aymane Ahajjam, Walker McKee, Sheridan Parker, Matthew Johnson, [Aaron Bergstrom](#) et al.
doi: [10.1016/j.acags.2026.100322](#)
- 2025 **Assessing the suitability of sites near Pine Island Glacier for subglacial bedrock drilling aimed at detecting Holocene retreat–readvance**, *The Cryosphere*
[Joanne S. Johnson](#), [John Woodward](#), [Ian M. Nesbitt](#), [Kate Winter](#), [Seth W. Campbell](#), [Kier Nichols](#), [Ryan Venturelli](#), [Scott Braddock](#), [Brent Goehring](#), [Brenda L. Hall](#) et al.
doi: [10.5194/tc-19-303-2025](#)
- 2025 **Miocene and Pliocene ice and air from the Allan Hills blue ice area, East Antarctica**, *Proceedings of the National Academy of Sciences*
[Sarah Shackleton](#), [Valens Hishamunda](#), L. Davidge, [Ed Brook](#), J. Marks-Peterson, [Austin Carter](#), S. Aarons, [Andrei Kurbatov](#), [Douglas Introne](#), Y. Yan, [Ian M. Nesbitt](#) et al.
doi: [10.1073/pnas.2502681122](#)
- in review **The sediment delivery continuum from deglaciation to the modern watershed based on lake sedimentary deposits in the Northeastern USA**, *Quaternary Research*
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Sean M.C. Smith](#), [Bess G. Koffman](#), [Steven A. Arcone](#), and [Kristin M. Schild](#)
preprint: [10.13140/RG.2.2.11567.05281](#)
- 2021 **rsudp: A Python package for real-time seismic monitoring with Raspberry Shake instruments**, *Journal of Open Source Software*
[Ian M. Nesbitt](#), Richard I. Boaz, and Justin Long
doi: [10.21105/joss.02565](#)
- 2020 **Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures**, *Science*
Thomas Lecocq, Stephen P Hicks, Koen Van Noten, Kasper van Wijk, Paula Koelemeijer, Raphael SM De Plaen, Frédéric Massin, Gregor Hillers, Robert E Anthony, Maria-Theresia Apoloner, ... [Ian M. Nesbitt](#) et al.
doi: [10.1126/science.abd2438](#)
- 2016 **Tracing subarctic Pacific water masses with benthic foraminiferal stable isotopes during the LGM and late Pleistocene**, *Deep-Sea Research Part II: Topical Studies in Oceanography*
[Mea S. Cook](#), A. Christina Ravelo, Alan C. Mix, [Ian M. Nesbitt](#), and Nari V. Miller
doi: [10.1016/j.dsr2.2016.02.006](#)

Conference proceedings, presentations, and datasets

- 2025 **Validation and interoperability of schema.org as a science metadata standard: a DataONE perspective**, *Earth Science Information Partners (ESIP) Annual Meeting*, Seattle, WA, USA, Invited talk
[Ian M. Nesbitt](#)
[Speaker page](#)
- 2022 **Post-glacial sediment delivery continuum to an impounded valley reach in central Maine: a multi-disciplinary approach**, *Geological Society of America Abstracts with Programs*, Invited talk
[Ian M. Nesbitt](#), [Sean M.C. Smith](#), [Seth W. Campbell](#), [Bess G. Koffman](#), [Steven A. Arcone](#), and [Kristin M. Schild](#)
doi: [10.1130/abs/2022NE-374798](#)
- 2021 **Unprecedented present deglaciation of Pine Island Glacier Inferred from Ice Penetrating Radar Studies of Localised Ice Domes in the Hudson Mountains**, *International Thwaites Glacier Collaboration Conference*, Poster
[John Woodward](#), [Joanne S. Johnson](#), [Seth W. Campbell](#), [Ian M. Nesbitt](#), [Brenda L. Hall](#), [Scott Braddock](#), [Meghan Spoth](#), [Brent Goehring](#), [Ryan Venturelli](#), [Dylan H. Rood](#), [Kier Nichols](#), [Johnathan Adams](#), and [Greg Balco](#)
- 2019 **A decision-making framework for sedimentation analyses in dammed river corridor impoundments**, *Geological Society of America Abstracts with Programs 51*, Portland, ME, Talk
[Ian M. Nesbitt](#), [Sean M.C. Smith](#), [Bess G. Koffman](#), [Seth W. Campbell](#), and [Steven A. Arcone](#)
doi: [10.1130/abs/2019NE-328587](#)
- 2018 **Sedimentary architecture and accumulation rates of multiple lakes in New England, USA**, *AGU Fall Meeting Abstracts 2018*, Washington, D.C., Poster NS41B-0830
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), [Sean M.C. Smith](#), and [Bess G. Koffman](#)
- 2018 **Sedimentary architecture of Maine lakes derived with ground-penetrating radar**, *Maine Water and Sustainability Conference*, Augusta, ME, Poster
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)
- 2018 **Holocene sediment volume determined by ground-penetrating radar and sidescan sonar in Maine, USA**, *Geological Society of America, Northeastern Section - 53rd Annual Meeting*, Talk
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)
doi: [10.1130/abs/2018NE-311370](#)
- 2017 **Using ground-penetrating radar and sidescan sonar to compare lake bottom geology in New England**, *AGU Fall Meeting Abstracts 2017*, New Orleans, LA, Talk PP44B-01
[Ian M. Nesbitt](#), [Seth W. Campbell](#), [Steven A. Arcone](#), and [Sean M.C. Smith](#)

- 2017 **New England lake bottom geology and sedimentation rates derived from ground-penetrating radar**, *GSA Annual Meeting*, Seattle, WA
Seth W. Campbell, Ian M. Nesbitt, Steven A. Arcone, Sean M.C. Smith
doi: [10.1130/abs/2017AM-306235](https://doi.org/10.1130/abs/2017AM-306235)
- 2015 **Geophysical imaging of dams and levees in the Northeast US**, *Society of Exploration Geophysicists Abstracts With Programs*, New Orleans, LA
Daniel A. Rosales, William F. Murphy IV, Matthew B. Art, Ian M. Nesbitt, Salvatore Triano, David C. Herron, Kurt Schollmeyer, James Trotta, W. Bruce Ward, Lisa Stewart, and William F. Murphy III
- 2014 **Tracing Bering Sea circulation with benthic foraminiferal stable isotopes during the Pleistocene**, *AGU Fall Meeting Abstracts 2014*, PP23D-08
Mea S. Cook, A. Christina Ravelo, Alan C. Mix, Ian M. Nesbitt, and Nari V. Miller
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Thesis, Department of Geosciences, Williams College*, Williamstown, MA
Ian M. Nesbitt
Advisor: David P. Dethier
Available via [Williams College Libraries](#)
doi: [10.13140/RG.2.2.33197.20966](https://doi.org/10.13140/RG.2.2.33197.20966)
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Geological Society of America, Northeastern Section - 48th Annual Meeting*, Bretton Woods, NH
Ian M. Nesbitt and David P. Dethier
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Twenty-sixth annual Keck Research Symposium in Geology, proceedings 2013*, Pomona, CA
Ian M. Nesbitt and Robert Varga

Open-source software

- 2025 **MetacatUI 2.36.0**, *Arctic Data Center*
Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
<https://github.com/NCEAS/MetacatUI>
doi: [10.18739/a2hx15s97](https://doi.org/10.18739/a2hx15s97)
- 2025 **MetacatUI 2.35.0**, *Arctic Data Center*
Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
<https://github.com/NCEAS/MetacatUI>
doi: [10.18739/a2q52ff87](https://doi.org/10.18739/a2q52ff87)

- 2024 **HashStore — Python**, *Arctic Data Center*
 DouMing Mok, Matthew Brooke, Jing Tao, Jeanette Clark, Matt Jones, and Ian M. Nesbitt
<https://github.com/DataONEorg/hashstore>
 doi: 10.18739/a2zg6g87q
- 2024 **MetacatUI 2.34.0**, *Arctic Data Center*
 Matt Jones, Chris Jones, Lauren Walker, Robyn Thiessen-Bock, Ben Leinfelder, Peter Slaughter, Bryce Mecum, Rushiraj Nenuji, Hasham Elbashandy, Val Hendrix et al.
<https://github.com/NCEAS/MetacatUI>
 doi: 10.18739/a2w669b2r
- 2021 **SeidarT: Seismic and radar survey modeling toolbox**
 Steven P. Bernsen, Christopher C. Gerbi, Ian M. Nesbitt, Ann Hill, Senthil Vel, Knut Christianson, Seth W. Campbell, and Ben Hills
<https://github.com/UMainedynamics/SeidarT>
 doi: 10.5281/zenodo.5498194
- 2020 **rsudp: Continuous visual display, sudden motion monitoring, and historical replay of Raspberry Shake data**
 Ian M. Nesbitt, Richard I. Boaz, and Justin Long
<https://github.com/raspishake/rsudp>
 doi: 10.21105/joss.02565
- 2019 **gpx2dzg: a tool to convert GPX files to GSSI's proprietary DZG format, and plot comparisons of marks in GPX and DZT/DZX files**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/gpx2dzg>
 doi: 10.5281/zenodo.3260948
- 2019 **readgssi: an open-source tool to read and plot GSSI ground-penetrating radar data**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/readgssi>
 doi: 10.5281/zenodo.1439119

Technical Skills

Languages	Python, R, Matlab, Node.js, Java, Ruby/Rails
Markup	Django Templates, HTML, CSS, $\text{\LaTeX}$, Markdown, Overleaf
Platform Tools	Docker, Kubernetes, Selfhosted Linux servers, Raspberry Pi, Heroku, AWS
Developer Tools	GNU/Linux, Unix Terminal/bash/zsh, GNU Make, git, GitHub, Jupyter Notebooks, Visual Studio Code, Apple Developer Tools
Bug Tracking	Request Tracker, MantisBT, GitHub
APIs	DataONE API, FDSN, USGS Earthquakes/EarthExplorer, Telegram, Google Maps, Google Workspace
Other Tools	QGIS, ArcGIS, Inkscape, GIMP, Illustrator, Photoshop, ReflexW, RADAN

Communication	Slack, Discord, Mattermost
Office Suites	Microsoft Office, LibreOffice, Google Docs/Sheets/Slides

Spoken Languages

English	Native
Spanish	Beginner

Certifications

Since 2022	SafeSport Trained
2013, 2018	Wilderness First-Aid
2016	CPR, First-Aid, and Defibrillator
2019	Antarctic Deep Field Survival
2020	T3 Alliance certified instructor